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Every year, American high school students take SATs, which are standardized tests intended to measure literacy, numeracy, and writing skills. There are three sections - reading, math, and writing, each with a **maximum score of 800 points**. These tests are extremely important for students and colleges, as they play a pivotal role in the admissions process.

Analyzing the performance of schools is important for a variety of stakeholders, including policy and education professionals, researchers, government, and even parents considering which school their children should attend.

You have been provided with a dataset called `schools.csv`, which is previewed below.

You have been tasked with answering three key questions about New York City (NYC) public school SAT performance.

```
# Re-run this cell
import pandas as pd

# Read in the data
schools = pd.read_csv("schools.csv")

# Preview the data
schools.head()

# Start coding here...
# Add as many cells as you like...
```

...	↑↓	school_name	...	↑↓	b...	...	↑↓	buildin...	...	↑↓	avera...	...	↑↓	average_r...	...	↑↓	average_w...	...
3		High School for Dual Language and Asian Stud...			Manhattan			M445			613			453				
1		Essex Street Academy			Manhattan			M445			395			411				
2		Lower Manhattan Arts Academy			Manhattan			M445			418			428				
4		Henry Street School for International Studies			Manhattan			M056			410			406				
0		New Explorations into Science, Technology an...			Manhattan			M022			657			601				

Rows: 5

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```
best_math_result = 0.8*800
best_math_schools = schools[schools['average_math']>=best_math_result][["school_name", 'average_math']]
best_math_schools = best_math_schools.sort_values(by='average_math', ascending=False)
# best_math_schools
```

```
# top_10_schools = schools.sort_values(by='percent_tested', ascending=False).head(10)
top_10_schools = schools
top_10_schools['total_SAT'] = top_10_schools['average_math'] + top_10_schools['average_reading'] +
top_10_schools['average_writing']

top_10_schools = top_10_schools[['school_name', 'total_SAT']].sort_values(by='total_SAT', ascending=False).head(10)
# top_10_schools
```

```
largest_std_dev = schools
largest_std_dev['total_SAT'] = largest_std_dev['average_math'] + largest_std_dev['average_reading'] +
largest_std_dev['average_writing']

largest_std_dev = largest_std_dev.groupby('borough').agg(num_schools = ('school_name', 'count'), average_SAT = ('total_SAT',
'mean'), std_SAT = ('total_SAT', 'std') )

largest_std_dev = largest_std_dev.round(2)
largest_std_dev = largest_std_dev.sort_values(by='std_SAT', ascending=False).head(1)

# largest_std_dev
```